

FitBuddy

AI Fitness Plan Generator Using Gemini Models

Reviewed & Presentation-Ready Architecture Overview

Slide 1 – Application Architecture Overview

- Activity 1.2 – Define the Architecture of the Application
- FitBuddy uses a 3-tier architecture to separate the user interface, application logic, and data management.
- Presentation Layer – User interface and user interaction
- Application Layer – Business logic, request processing, authentication and API communication
- Data Layer – Database, CRUD operations, consistency and data management
- Separation of concerns makes the system easier to maintain, secure and scale.
- Gemini models can support AI-driven fitness-plan generation within the application layer.

Architecture at a glance: Presentation Layer ↔ Application Layer ↔ Data Layer

Slide 2 – Presentation Layer

- Purpose: Provides the main interaction between the user and the FitBuddy application.
- Components: User Interface (UI), Login/Registration pages, Dashboard, Forms and input screens, Reports and notifications.
- Responsibilities: Accept user input and display generated results.
- Provide simple, user-friendly navigation and basic input validation.
- Present fitness-plan information clearly so users can understand the generated recommendations.

Slide 3 – Application & Data Layers

- Application Layer: Processes user requests and implements application business rules.
- Handles authentication and authorization and coordinates communication between the UI, AI service/API and database.
- AI integration: Gemini models can be called from the application layer to generate personalized fitness-plan content based on the application's inputs.
- Data Layer: Stores application data and manages database operations.
- Performs Create, Read, Update and Delete (CRUD) operations while maintaining data consistency and security.
- Typical backend flow: Backend/API → Business Logic → Gemini/API service and/or Database.

Slide 4 – Overall Architecture & Data Flow

- Architecture Flow: User → User Interface → Application/API → Gemini/Business Logic → Database (as required) → Application → UI
- 1. User submits fitness-related inputs through the interface.
- 2. Application layer validates and processes the request.
- 3. Business logic prepares the request for the AI service when an AI-generated plan is required.
- 4. Gemini model returns generated plan content to the application.
- 5. The application can store or retrieve relevant data through the database.
- 6. The processed result is returned to the UI and displayed to the user.
- Key benefits: Modular design, easier maintenance, better separation of responsibilities, scalability, security and efficient data management.

Slide 5 – Conclusion

- FitBuddy's 3-tier architecture provides a clear separation between presentation, application processing and data management.
- The architecture supports a structured integration of Gemini models for AI-assisted fitness-plan generation.
- Modular components make the application easier to maintain and extend.
- Clear data flow helps keep user interaction, business logic, AI processing and database operations organized.
- Overall, the architecture provides a strong foundation for developing and scaling the FitBuddy application.